

Lecture 17

Dressing, excitations and the thermodynamic Bethe ansatz

Objectives. Course map: P4, and G3 completed. [Lecture 16](#) put the ground state into the thermodynamic limit. This lecture dresses it and heats it. One linear operator, dressing, turns the bare energy of an added particle into its cost and the derivative of its bare momentum into the derivative of its momentum, and gives the gas its two branches of excitations and its speed of sound. Above zero temperature, counting the ways to fill the vacancies gives an entropy, and minimising the free energy against it gives one nonlinear integral equation, the Yang–Yang equation, whose solution is the gas’s exact equation of state.

17.1 The question

[Lecture 16](#) found the ground states. Physics happens above them. So the question is

what are the excitations, and what is the equation of state at any temperature?

The answer: dressing ([Definition 17.1](#)) gives an excitation’s energy and the derivative of its momentum ([Proposition 17.2](#)), which gives Lieb’s two branches; at $T > 0$ the Yang–Yang equation ([Theorem 17.7](#)) gives the pressure. The worked example is the gas at $T = 0$.

17.2 Dressing

Every question asked of a Bethe state in the thermodynamic limit has the same shape. Adding one particle to the sea moves *every other* root a little, so the change in any extensive quantity is the particle’s own contribution *plus* the backflow. One linear operator does the bookkeeping for all of them.

Definition 17.1 (Dressing). Let $\vartheta(\lambda) = \rho(\lambda)/\rho_t(\lambda)$ be the filling fraction, $\rho_t = \rho + \rho_h$ (at $T = 0$, the indicator of the sea). The dressed f is the solution of

$$f^{\text{dr}}(\lambda) = f(\lambda) + \frac{1}{2\pi} \int K(\lambda - \mu) \vartheta(\mu) f^{\text{dr}}(\mu) d\mu, \quad (17.1)$$

with K the model’s kernel ([16.6](#)).

The definition is worth its own name because ([16.5](#)) is already an instance of it. **It is a response, not a total:** the dressed bare momentum k_0^{dr} is not the excitation’s momentum; the momentum comes from integrating $(k'_0)^{\text{dr}}$.

Proposition 17.2. $2\pi\rho_t = (p'_0)^{\text{dr}}$ for any ϑ . Now take $T = 0$, so that ϑ is the indicator of the sea, and let an excitation have bare energy ε_0 and bare momentum k_0 ; write $\varepsilon = \varepsilon_0^{\text{dr}}$. If ε vanishes at the edges of the sea, then $\varepsilon' = (\varepsilon'_0)^{\text{dr}}$, and the excitation's dressed velocity is

$$v^{\text{eff}}(\lambda) = \frac{(\varepsilon'_0)^{\text{dr}}(\lambda)}{(k'_0)^{\text{dr}}(\lambda)} = \frac{d\varepsilon}{dp}, \quad p' = (k'_0)^{\text{dr}}. \quad (17.2)$$

Proof (★). Multiply (16.5) by 2π and use $\rho = \vartheta\rho_t$: $2\pi\rho_t = p'_0 + \frac{1}{2\pi} \int K\vartheta(2\pi\rho_t)$, which is (17.1) for $f = p'_0$, for any ϑ . For the second statement, differentiate (17.1) in λ and integrate by parts in μ , using $\partial_\lambda K(\lambda - \mu) = -\partial_\mu K(\lambda - \mu)$:

$$\varepsilon'(\lambda) = \varepsilon'_0(\lambda) + \frac{1}{2\pi} \int K(\lambda - \mu)(\vartheta\varepsilon)'(\mu)d\mu.$$

ϑ being an indicator, $(\vartheta\varepsilon)'$ is $\vartheta\varepsilon'$ plus delta functions at the edges weighted by ε there. Those vanish exactly when ε does, and what is left is (17.1) for ε'_0 . The velocity is then the chain rule. *Without the edge condition the boundary terms survive*, and $\varepsilon' \neq (\varepsilon'_0)^{\text{dr}}$: for the gas at $c = q = 1$, dressing $f(k) = k$ and differentiating gives 1.2099 at $k = 0$ against $(1)^{\text{dr}}(0) = 1.9190$. *The step also needs $T = 0$* : for a smooth ϑ the extra term is $\frac{1}{2\pi} \int K\vartheta'\varepsilon$, a bulk integral that no edge condition kills. ■

Remark 17.3 (Which momentum, and which way it runs). k_0 is the excitation's own bare momentum, not automatically (16.5)'s counting momentum p_0 . For a particle added to the gas the two agree; for the chain they do not, since $p_0 = \theta_1$ runs backwards against $k = \pi - \theta_1$ (Lecture 16) — and the chain's excitation is not a magnon but a *hole*, of bare energy $-\varepsilon_0$, whose momentum runs *with* ζ . The spinon reading at the end computes it and finds $|v^{\text{eff}}| \rightarrow \pi J/2$; reading (17.2) with ε_0 and θ_1 instead flips that sign.

Two things fix the object in place. Equation (17.1) is *linear* with the same kernel for every f , so dressing is one operator $(1 - \frac{1}{2\pi}K\vartheta)^{-1}$: a single numerical solve answers every question. And ϑ carries all the state dependence — sharp at $T = 0$, a smooth Fermi function at finite temperature (below), position- and time-dependent out of equilibrium (Lecture 20) — with (17.2) the dressed velocity throughout. What does *not* carry over is the route to it: at $T > 0$ the identity $\varepsilon' = (\varepsilon'_0)^{\text{dr}}$ comes from differentiating Yang–Yang's *nonlinear* equation for the pseudo-energy (Corollary 17.8).

17.3 Lieb's two branches

The gas has a Fermi sea with *edges*, so it has two ways to be excited; Lieb found both in 1963, and both come off one pair of functions. The $T = 0$ dressed energy and momentum are

$$\varepsilon(k) - \frac{1}{2\pi} \int_{-q}^q K(k - k')\varepsilon(k')dk' = k^2 - \mu, \quad p'(k) = 2\pi\rho_t(k), \quad (17.3)$$

the first being (17.1) for the bare energy $k^2 - \mu$ and the second Proposition 17.2; the constant μ — the *chemical potential*, and in this lecture there is no spectral parameter to confuse it with — is fixed by $\varepsilon(\pm q) = 0$. That condition does two things at once: it puts both branches at zero energy where they begin, and it is exactly the hypothesis of Proposition 17.2, so the velocity formula applies. Now add a particle just above the right edge, $k > q$: energy $\varepsilon(k) > 0$ and momentum $p(k) - p(q)$ trace the *Type I* branch, the Bogoliubov mode (a particle

below $-q$ gives the mirror branch, referred to $-q$). Move a particle from inside the sea to its edge, leaving a hole at $|k| < q$: energy $-\varepsilon(k) > 0$ and momentum $p(q) - p(k)$ trace *Type II*, which rises, turns over and returns to zero at momentum $2\pi n$ · *sketched: the branches are computed here from (17.3); that they exhaust the elementary excitations is cited (Lieb 1963).*

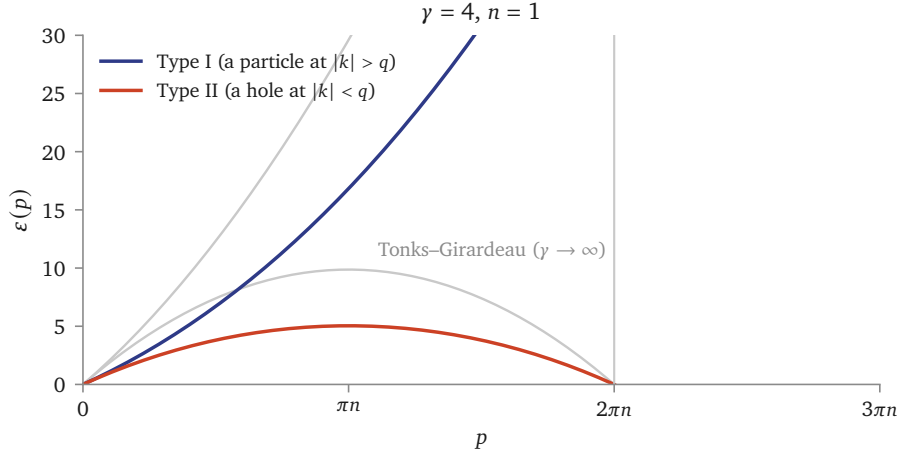


Figure 17.1: Lieb's branches at $\gamma = 4$, $n = 1$. Both leave the origin with the same slope — the dressed velocity v^{eff} (17.2) at the Fermi edge, i.e. the speed of sound. The Type II branch vanishes again at $p = 2\pi n$: the sea has been rotated by one unit. Faint: the Tonks–Girardeau limits $(p + \pi n)^2 - (\pi n)^2$ and $(\pi n)^2 - (\pi n - p)^2$ that the two branches approach.

Type II has no counterpart in Bogoliubov theory — it is the exact version of a dark soliton · *cited: Kulish–Manakov–Faddeev (1976)* — and both branches are linear at small momentum with the same slope, the sound velocity that makes the gas a Luttinger liquid.

17.4 Counting

Everything in this lecture rests on one sentence of [Lecture 16](#): a Bethe state is not a list of rapidities but a choice of *quantum numbers* from the list of *vacancies*. Choose a subset, solve (16.1), and the roots come out. So counting states is counting subsets — and that is a problem with an answer. For the chain the list is finite, of length $N - M$, and *on the branch where ζ increases* the roots occupy M distinct members of it ([Proposition 16.1](#), whose second sentence [Remark 16.2](#) showed to be a hypothesis); for the gas the list is all of $\mathbb{Z} + \frac{N-1}{2}$ and is not finite at all. What the count below uses is neither, but the local *density* of vacancies ρ_t , which both models have.

Fix a coarse-graining: cut the rapidity line into cells of width $d\lambda$, small on the scale over which ρ varies but wide enough to hold many vacancies. In the cell at λ there are $L\rho_t(\lambda)d\lambda$ slots and $L\rho(\lambda)d\lambda$ of them carry roots, where L is the number of sites N or the length of the box, as in (16.5). The cells are independent, so

Proposition 17.4 (The Yang–Yang entropy). *The number of Bethe states compatible with the densities ρ, ρ_h is $\prod_{\text{cells}} \binom{L\rho_t d\lambda}{L\rho d\lambda}$, and its logarithm is $Ls + o(L)$ with*

$$s = \int [\rho_t \ln \rho_t - \rho \ln \rho - \rho_h \ln \rho_h] d\lambda = \int \rho_t [-\vartheta \ln \vartheta - (1 - \vartheta) \ln(1 - \vartheta)] d\lambda, \quad (17.4)$$

the entropy per site (or per unit length), with $\vartheta = \rho/\rho_t$ the filling fraction of [Definition 17.1](#).

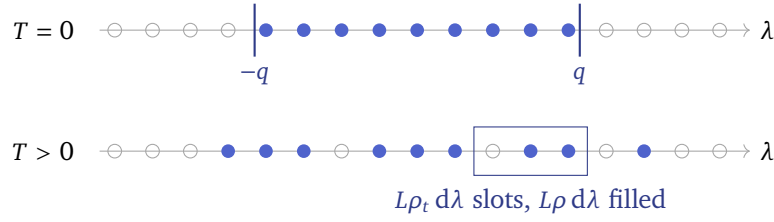


Figure 17.2: A Bethe state is a choice of which vacancies carry roots. At $T = 0$ the choice is forced — a solid block: no holes inside it, no roots outside (Lecture 16). At $T > 0$ any subset is allowed, and a cell of the rapidity line with $L\rho_t d\lambda$ slots and $L\rho d\lambda$ of them filled can be filled in $\binom{L\rho_t d\lambda}{L\rho d\lambda}$ ways. The logarithm of the product over cells is the Yang–Yang entropy.

· proved as a count, from the hypotheses named below

Proof (★). Stirling gives $\ln \binom{A}{B} = A \ln A - B \ln B - (A - B) \ln(A - B) + O(\ln A)$, so *three* limits are needed and their order is not free. Cut the rapidity line at $|\lambda| \leq \Lambda$ and take finitely many cells there. Let $L \rightarrow \infty$: the leading terms are $L d\lambda$ times the bracket in (17.4) and the errors are $O(\ln L)$ in each of finitely many cells, hence $o(L)$. Then refine the cells, which turns the sum into the integral. Then let $\Lambda \rightarrow \infty$, which converges because the entropy density is integrable. Taking the first two in the other order would not do — with one cell per vacancy the errors are $O(L \ln L)$ — and without the cutoff the first step never starts, since the gas’s outermost cell holds infinitely many slots. The second form is the first with $\rho = \vartheta \rho_t$, $\rho_h = (1 - \vartheta) \rho_t$. ■

Remark 17.5 (What the count assumes). For the gas it assumes, first, that *every* way of filling the slots is admissible, with no further extensive restriction within a cell or between cells; second, that distinct choices give distinct states and exhaust the spectrum — that quantum numbers and solutions are in bijection is Proposition 11.5, and that the states so built are all of them is cited; third, that all choices with the same coarse-grained densities are equally numerous. None of this is proved here. The variational identities below are proved given this count; the saddle-point identification is sketched, and uniqueness is cited. The chain needs a string hypothesis as well (the reading at the end).

17.5 Minimising the free energy

At temperature T and chemical potential μ the equilibrium state minimises the grand free energy per unit length,

$$g[\rho] = \int (\varepsilon_0(\lambda) - \mu) \rho(\lambda) d\lambda - Ts, \quad (17.5)$$

where ε_0 is the bare energy of one root — $\varepsilon_0 = k^2$ for the gas of Lecture 11, $\varepsilon_0 = -\pi J a_1$ for the chain (the reading at the end). Here μ is the *chemical potential*, as in Lieb’s branches.

Remark 17.6 (ρ_h is not yours to vary). Equation (17.5) is a functional of ρ alone. The hole density is fixed by (16.5),

$$\rho + \rho_h = \frac{p'_0}{2\pi} + \frac{1}{2\pi} \int K(\lambda - \lambda') \rho(\lambda') d\lambda', \quad \text{so} \quad \delta \rho_h = -\delta \rho + \frac{1}{2\pi} \int K \delta \rho. \quad (17.6)$$

The kernel term is the whole of the interaction. Drop it and the variation returns free fermions with the bare dispersion; treat ρ_h as an independent function and the answer is a Boltzmann factor with no exclusion at all. Both are checked in the script, and the second is the error the section is named after.

Theorem 17.7 (Yang–Yang). Write $\varepsilon = T \ln(\rho_h/\rho)$, the pseudo-energy. Then ρ is stationary for (17.5) subject to (17.6) if and only if

$$\varepsilon(\lambda) = \varepsilon_0(\lambda) - \mu - \frac{T}{2\pi} \int K(\lambda - \lambda') \ln(1 + e^{-\varepsilon(\lambda')/T}) d\lambda', \quad (17.7)$$

and then the filling fraction is a Fermi function of the pseudo-energy, and the minimum value of g is

$$\vartheta = \frac{1}{1 + e^{\varepsilon/T}}, \quad \min g = -\frac{T}{2\pi} \int p'_0(\lambda) \ln(1 + e^{-\varepsilon(\lambda)/T}) d\lambda. \quad (17.8)$$

For the gas $p'_0 = 1$ and $-\min g$ is the pressure P .

· proved from [Proposition 17.4](#); that the minimiser gives the true free energy is Yang and Yang's saddle-point argument, sketched after the proof

Proof. From (17.4), $\delta s = \int [\ln(\rho_t/\rho) \delta \rho + \ln(\rho_t/\rho_h) \delta \rho_h]$, and $\rho_h/\rho = e^{\varepsilon/T}$ turns the two logarithms into $\ln(1 + e^{\varepsilon/T})$ and $\ln(1 + e^{-\varepsilon/T})$. Substituting $\delta \rho_h$ from (17.6) and using that K is even,

$$\frac{\delta g}{\delta \rho(\lambda)} = (\varepsilon_0 - \mu) - T \left[\ln(1 + e^{\varepsilon/T}) - \ln(1 + e^{-\varepsilon/T}) \right] - \frac{T}{2\pi} \int K(\lambda - \lambda') \ln(1 + e^{-\varepsilon(\lambda')/T}) d\lambda'.$$

The bracket is ε/T identically, so the vanishing of the derivative is (17.7). The first equation of (17.8) is the definition of ε rearranged. For the second, write $Ts = \int \rho \varepsilon + T \int \rho_t \ln(1 + e^{-\varepsilon/T})$, use (17.7) to replace $\varepsilon_0 - \mu$ in (17.5), and use (17.6) to replace ρ_t : the two double integrals are equal by the symmetry of K and cancel, leaving $g = -\frac{T}{2\pi} \int p'_0 \ln(1 + e^{-\varepsilon/T})$. The weight p'_0 is not decoration — it is what the chain's strings need (the reading at the end), and dropping it would make the chain's free energy diverge. Finally the stationary point is a minimum: writing $\sigma(\vartheta) = -\vartheta \ln \vartheta - (1 - \vartheta) \ln(1 - \vartheta)$, the integrand of (17.4) is $\rho_t \sigma(\rho/\rho_t)$, the perspective of the concave σ and therefore jointly concave in (ρ, ρ_t) ; and $\rho \mapsto (\rho, \rho_t[\rho])$ is affine by (17.6). So s is concave in ρ , g is convex, and the admissible set $\{0 \leq \rho \leq \rho_t[\rho]\}$ is convex. That (17.7) has exactly one solution at every $c > 0$, $T > 0$ is Yang and Yang's, and is cited. ■

Why the minimum is the free energy. The argument is a saddle point. Write the partition function as a sum over sets of quantum numbers, group the sets by their coarse-grained densities, and the sum becomes $Z = \sum_{\{\rho\}} e^{-Lg[\rho]/T}$ over the coarse-grained configurations, with g of (17.5): the number of terms in each group is e^{Ls} by [Proposition 17.4](#) and each contributes $e^{-L(e-\mu n)/T}$. As $L \rightarrow \infty$ the sum is dominated by its largest term and $-T \ln Z/L \rightarrow \min g$, provided the number of groups grows sub-exponentially — which is the usual requirement, and the usual thing that is waved through.

Equation (17.7) is nonlinear — the unknown sits inside a logarithm inside the integral — and that is the price of the temperature. Everything else about it is a gift: one scalar equation, one scalar unknown, and no reference to N or to any particular state. And what the second half of (17.8) says is that the minimisation is a Legendre transform. The minimum value of

$g = e - \mu n - Ts$ is the grand potential density, $-P(\mu, T)$; so P is the Legendre transform of the energy density in both n and s , its partners come back by differentiating,

$$n = \frac{\partial P}{\partial \mu}, \quad s = \frac{\partial P}{\partial T}, \quad \frac{E}{L} = Ts + \mu n - P, \quad (17.9)$$

and one scalar function of two variables carries the whole thermodynamics. Students lose the last of these by trying to compute E/L from $\int \varepsilon \rho$, which is not it: ε is not an energy.

Corollary 17.8 (Dressing at finite temperature). $\varepsilon' = (\varepsilon'_0)^{\text{dr}}$ at every temperature, with ϑ of (17.8) in (17.1), and $2\pi\rho_t = (p'_0)^{\text{dr}}$. The effective velocity at temperature T is defined, for the gas, by the same ratio as at $T = 0$,

$$v^{\text{eff}}(k) = \frac{\varepsilon'(k)}{2\pi\rho_t(k)}, \quad (17.10)$$

which reduces to (17.2) when ϑ is an indicator. For the chain the denominator must be the excitation's own bare momentum dressed, not the counting momentum's (Remark 17.3).

Proof. Differentiate (17.7) in λ , move the derivative onto μ' by $\partial_\lambda K(\lambda - \lambda') = -\partial_{\lambda'} K(\lambda - \lambda')$ and integrate by parts; the boundary terms vanish because $K \rightarrow 0$. What is left involves $\partial_{\lambda'} \ln(1 + e^{-\varepsilon'/T}) = -\vartheta \varepsilon'/T$, and the T cancels:

$$\varepsilon'(\lambda) = \varepsilon'_0(\lambda) + \frac{1}{2\pi} \int K(\lambda - \lambda') \vartheta(\lambda') \varepsilon'(\lambda') d\lambda',$$

which is (17.1) for ε'_0 . The second statement is Proposition 17.2, whose first half holds for any ϑ ; the third is a definition. ■

17.6 Two checks

A new equation earns trust by reproducing the two answers already known.

Proposition 17.9 (Impenetrable, and cold). For the gas of Lecture 11, with $\varepsilon_0 = k^2$ and $K = 2c/(k^2 + c^2)$:

1. as $c \rightarrow \infty$, $\varepsilon(k) = k^2 - \mu - 2P/c + O(c^{-2})$ uniformly on compact sets of k , and P tends to the free-fermion pressure $\frac{T}{2\pi} \int \ln(1 + e^{-(k^2 - \mu)/T}) dk$, the correction being $O(1/c)$;
2. as $T \rightarrow 0$ at $\mu > 0$, (17.7) becomes (17.3) and $P \rightarrow -\frac{1}{2\pi} \int_{-q}^q \varepsilon = \mu n - E/L$, with the edge condition $\varepsilon(\pm q) = 0$ deduced, not imposed. (At $\mu < 0$ the limit is the vacuum: $\varepsilon = k^2 - \mu > 0$ everywhere, no sea, no zero, $n = P = 0$.)

Proof (★). Both parts need one imported fact: on any bounded set of parameters there is a C with $\varepsilon(k) \geq k^2 - C$, uniformly · cited: Yang and Yang (1969). It is what makes $\ln(1 + e^{-\varepsilon/T})$ integrable with bounded zeroth and second moments, and both proofs use exactly that.

(1) $\int K dk = 2\pi$ for every c , and $K(k) = \frac{2}{c} + O(c^{-3}k^2)$. With the moments bounded uniformly in $c \geq c_0$, for k in a compact set the integral in (17.7) is $\frac{T}{2\pi} \cdot \frac{2}{c} \int \ln(1 + e^{-\varepsilon/T}) + O(c^{-3}) = 2P/c + O(c^{-3})$; expanding P itself then leaves $O(c^{-2})$. That is a constant shift of μ . (2) $T \ln(1 + e^{-\varepsilon/T}) \rightarrow \max(0, -\varepsilon)$ pointwise and to within $T \ln 2$ uniformly — enough on a compact set, and the tails are handled by the same quadratic bound, *not* by the uniform bound, whose integral over $\mathbb{R}$ is infinite. The limit is supported where $\varepsilon < 0$; since ε is continuous and strictly increasing in $|k|$ · cited: Yang and Yang (1969), that set is an interval,

empty at $\mu < 0$ — the vacuum — and otherwise $(-q, q)$ with $\varepsilon(\pm q) = 0$. On it the limit of (17.7) is $\varepsilon = k^2 - \mu + \frac{1}{2\pi} \int_{-q}^q K \varepsilon$, which is (17.3). For P , split (17.8) the same way; $\mu n - E/L$ then follows from $P = Ts + \mu n - E/L$ and $s \rightarrow 0$. ■

Part (1) carries a sentence of physics, and it is the opposite of the one that suggests itself. The shift *raises* μ , so the gas has *more* pressure than the impenetrable one, while hard rods of length a give $P = P_F(\mu - aP)$ and less: matching, $a = -2/c$. At large finite c the gas is free fermions in a box *lengthened* by $2/c$ per particle. The Bethe equations say so directly — $\theta(k) \approx 2k/c$ turns (11.11) into $k_j(L + 2N/c) = 2\pi I_j$ in the rest frame, $\sum_l k_l = 0$ — and so does Lecture 16's $E/N = n^2 e = \frac{\pi^2 n^2}{3} (1 - 4/\gamma + \dots)$, which to this order is $\pi^2 n_{\text{eff}}^2/3$ with $n_{\text{eff}} = n/(1 + 2n/c)$. At $c = 100$, $T = \mu = 1$ the shift accounts for the whole of the 0.6% by which the interacting pressure exceeds the free-fermion one.

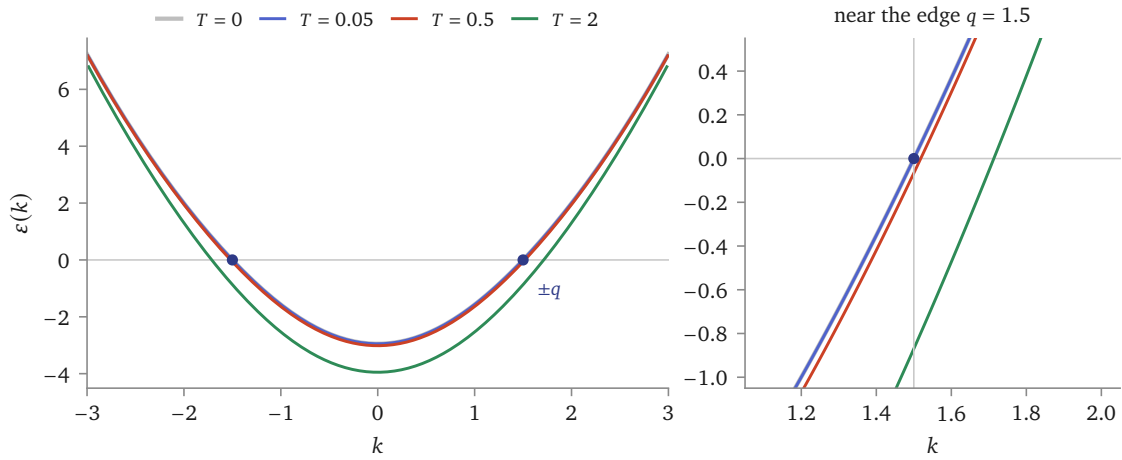


Figure 17.3: The pseudo-energy $\varepsilon(k)$ of (17.7) at three temperatures, $c = 1$ and μ held at the value (17.3) gives for $q = 1.5$. Grey: the $T = 0$ dressed energy (17.3), whose zeros are the Fermi edges $\pm q$. At $T > 0$ there are no edges: the curve slides down and $\varepsilon(\pm q)$ moves away from zero, which is why the second half of Proposition 17.2 cannot be used here; the finite-temperature identity follows instead from Corollary 17.8.

17.7 Solving it

Equation (17.7) has no closed-form solution and does not need one: it is a fixed-point equation, and the obvious iteration works. It also has one fewer parameter than it looks: substituting $k = \sqrt{T} \kappa$ and $c = \sqrt{T} \hat{c}$ makes every $\sqrt{T}$ cancel between K and dk' , so that ε/T and $P/T^{3/2}$ depend on $(\mu/T, \hat{c})$ alone. One plot is the whole equation of state.

Proposition 17.10 (The iteration contracts). *Let $\varepsilon \mapsto \mathcal{F}\varepsilon$ be the right side of (17.7). If $\vartheta \leq \bar{\vartheta}$ along the iteration then $\|\mathcal{F}\varepsilon - \mathcal{F}\tilde{\varepsilon}\|_\infty \leq \bar{\vartheta} \|\varepsilon - \tilde{\varepsilon}\|_\infty$.*

Proof (★). $\partial_\varepsilon \ln(1 + e^{-\varepsilon/T}) = -\vartheta/T$, and $\frac{1}{2\pi} \int |K| = 1$ in both models — the one place in this lecture where the sign of K (Lecture 16) does not matter. So the T cancels and the factor is $\sup \vartheta$. ■

The linearisation being differentiated here is nothing other than the dressing operator $\frac{1}{2\pi}K\vartheta$ of (17.1), so the equation that answers every question about the state is also the one that says how fast the solver converges.

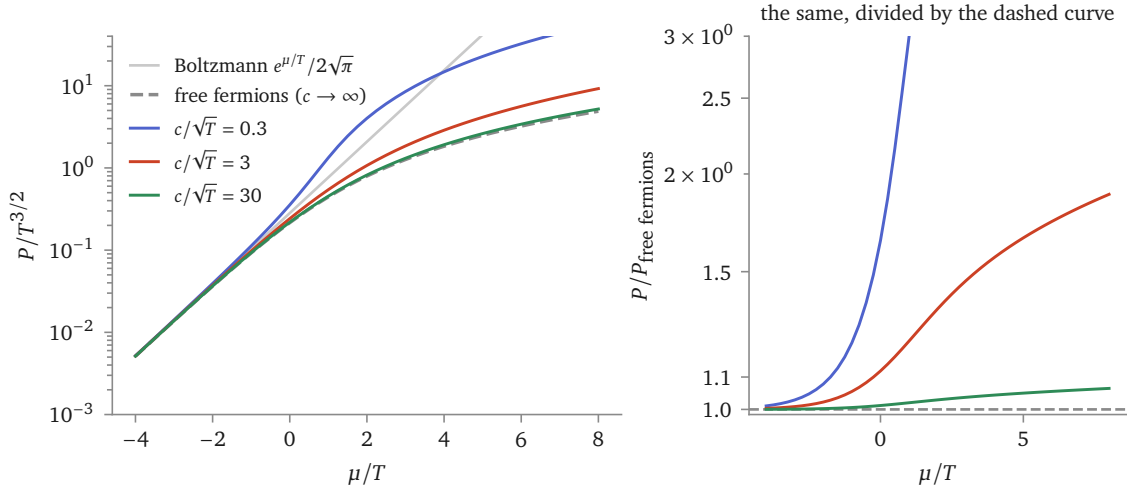


Figure 17.4: The equation of state. By scaling, (17.7) depends on $(\mu/T, c/\sqrt{T})$ only, so one plot of $P/T^{3/2}$ against μ/T carries every temperature. Dashed: the impenetrable limit $c \rightarrow \infty$, free fermions. Faint: the Boltzmann asymptote $e^{\mu/T}/2\sqrt{\pi}$, which every curve joins on the dilute side, interactions or not. The approach to the dashed curve is governed by $\gamma = c/n$, not by $c/\sqrt{T}$: at $\mu/T = 8$ the gas is degenerate and $c/\sqrt{T} = 30$ is still 7% above it.

Two features of Figure 17.4 deserve naming. On the dilute side every curve collapses onto $P/T^{3/2} = e^{\mu/T}/2\sqrt{\pi}$: a gas dilute enough is classical, and the coupling cannot be seen. On the degenerate side the curves fan out, and the free-fermion curve is the floor: since $K > 0$, (17.7) gives $\varepsilon \leq k^2 - \mu$ pointwise and hence $P \geq P_F$ at the same μ and T . The impenetrable gas has the least pressure there is, which is the statement of Proposition 17.9(1) seen from the other side.

Equation (17.7) is the equation of state of a real gas: a density profile measured on an atom chip in 2008 is reproduced with no free interaction parameter (the reading at the end).

17.8 What it buys

- Dressing, one linear operator for every quantity, and the dressed velocity (Proposition 17.2 and Corollary 17.8): the objects Lecture 20 carries out of equilibrium.
- Lieb's two branches and the speed of sound (Figure 17.1).
- The exact equation of state of an interacting quantum gas at every temperature and coupling (Theorem 17.7), checked in both limits (Proposition 17.9) and measured. Goal G3 is reached.

17.9 Caveats, in one place

- Status. Proposition 17.2, Theorem 17.7, Corollary 17.8, Proposition 17.9 and Proposition 17.10 are proved, the first, fourth and fifth in readings; Theorem 17.7 rests on the

count, whose assumptions (Remark 17.5) are not proved. That the minimum is the free energy is a saddle-point argument, sketched; that the Yang–Yang equation has one solution is cited, as is that Lieb’s two branches are all the elementary excitations.

- *Two hypotheses in Proposition 17.2.* Its second half needs $T = 0$ and $\varepsilon(\pm q) = 0$; at $T > 0$ the identity is derived again, from the Yang–Yang equation (Corollary 17.8).
- *The effective velocity at $T > 0$ is a definition (17.10);* that it is a propagation speed is not proved here.
- *The chain needs strings, and the string hypothesis is false as a statement about individual solutions; its thermodynamics is a reading.*

Reading

17.10 The chain’s excitations: spinons

Return to the antiferromagnet and make the cheapest excitation there is: remove one root. The sector is $M = N/2 - 1$, which by Proposition 16.1 has $N/2 + 1$ vacancies for $N/2 - 1$ roots — two holes, at $\lambda_{h_1}, \lambda_{h_2}$ — while the magnetisation has gone up by one, $\Delta S^z = 1$. Two objects, one unit of S^z between them: by itself that would also allow a spin 1 and a spin 0. What settles it is the multiplet structure — at fixed $\lambda_{h_1}, \lambda_{h_2}$ the states form one triplet and one singlet, exactly as two spin- $\frac{1}{2}$ objects do — cited: *Faddeev–Takhtajan (1981)*. They are the *spinons*, and that is the most surprising sentence about the chain: the elementary excitation of a spin chain is not a magnon of spin 1 but half of one, and it cannot be made alone.

Theorem 17.11 (The spinon). *Let N be even and let a state of the antiferromagnet lie on the branch of Proposition 16.1 — pairwise distinct real roots, ζ increasing — with $M = (N - r)/2$ of them, so that it carries r holes, necessarily an even number, and let r stay fixed as $N \rightarrow \infty$. In the limit of Theorem 16.3, if the holes sit at $\lambda_{h_1}, \dots, \lambda_{h_r}$ then*

$$\Delta E = \sum_i \varepsilon(\lambda_{h_i}), \quad \Delta K = \sum_i p(\lambda_{h_i}) \pmod{2\pi}, \quad (17.11)$$

with

$$\varepsilon(\lambda) = \frac{\pi J}{2 \cosh \pi \lambda}, \quad p(\lambda) = 2\pi \int_{-\infty}^{\lambda} \rho = \frac{\pi}{2} + \arctan \sinh \pi \lambda \in (0, \pi), \quad \varepsilon = \frac{\pi J}{2} \sin p. \quad (17.12)$$

• *proved from the first-order form of the assumption noted after the proof; the dispersion is that of des Cloizeaux and Pearson (1962)*

Proof. With ε_0 as above, $E = N \int \varepsilon_0 \rho$. With holes, (16.4) reads $(1 + a_2^*)\rho = a_1 - \rho_h$, while $(1 + a_2^*)\rho_{(0)} = a_1$ for the sea (16.8); subtracting and putting $\rho_h = \frac{1}{N} \sum_i \delta_{\lambda_{h_i}}$, $\rho - \rho_{(0)} = -\frac{1}{N}(1 + a_2^*)^{-1} \sum_i \delta_{\lambda_{h_i}}$. Hence, $1 + a_2^*$ being symmetric,

$$\Delta E = N \int \varepsilon_0 (\rho - \rho_{(0)}) = - \sum_i [(1 + a_2^*)^{-1} \varepsilon_0](\lambda_{h_i}) = \sum_i \varepsilon(\lambda_{h_i}), \quad (1 + a_2^*)\varepsilon = \pi J a_1,$$

which is (16.7) up to πJ : so $\varepsilon = \pi J / (2 \cosh \pi \lambda)$. For the momentum, (10.19) gives $K = \pi M - \frac{2\pi}{N} \sum_j I_j$ exactly, and deleting r roots from a symmetric filled set leaves $\sum_j I_j = - \sum_i I_{h_i}$, so $\Delta K = -\frac{r}{2}\pi + \frac{2\pi}{N} \sum_i I_{h_i}$, and $\frac{2\pi}{N} I_h = \zeta(\lambda_h) = \zeta(-\infty) + 2\pi \int_{-\infty}^{\lambda_h} \rho_t \rightarrow -\frac{\pi}{2} + p(\lambda_h)$, since $\zeta(-\infty) = -(N + r)\pi/2N \rightarrow -\pi/2$ at fixed r . Summing gives $\Delta K = -r\pi + \sum_i p(\lambda_{h_i})$, and r is even. Finally $p' = 2\pi \rho_{(0)} = \pi / \cosh \pi \lambda > 0$ integrates to (17.12), and $\sin p = 1 / \cosh \pi \lambda$. ■

The proof is a *backflow* computation. With the bare magnon energy $\varepsilon_0 = -2J/(4\lambda^2 + 1) = -\pi J a_1$, putting r delta-function holes into (16.4) and inverting $1 + a_2^*$ dresses the energy of the *removed* magnon, $-\varepsilon_0 = \pi J a_1$ — and that equation is (16.7) again, so $\varepsilon = (-\varepsilon_0)^{\text{dr}} = \pi J \rho$. It runs on *one order finer* information than Theorem 16.3's assumption supplies: at fixed r the hole density tends to zero, and (17.11) needs $\lim N(\rho_N - \rho_{(0),N})$. The careful route subtracts the two finite systems and linearises *before* letting $N \rightarrow \infty$ · cited: Takahashi §3.4; it gives the same ε , and finite chains agree (the check script, $N = 64$ to 1024).

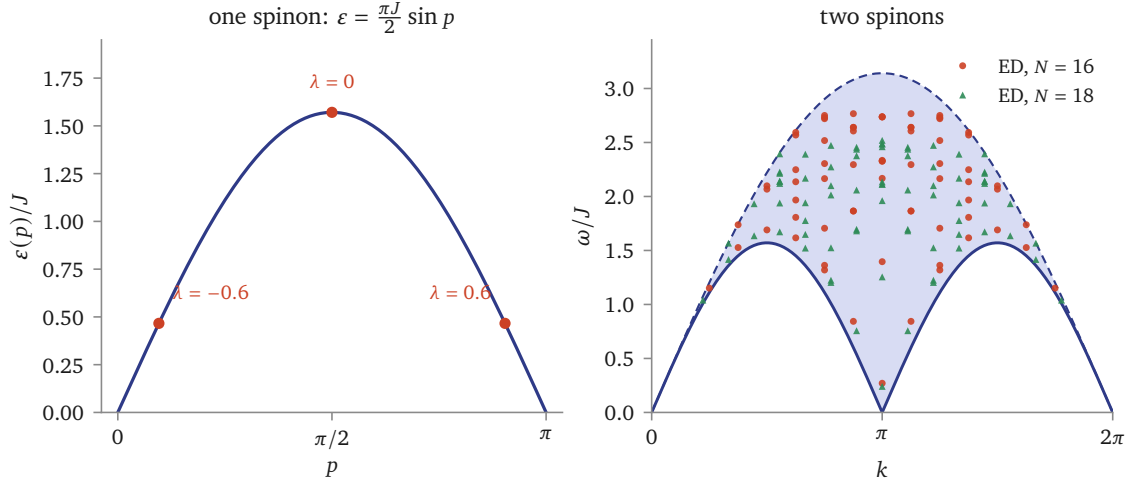


Figure 17.5: Left: the spinon dispersion $\varepsilon = \frac{\pi J}{2} \sin p$ of des Cloizeaux and Pearson, with three rapidities marked; $\lambda = 0$ sits at the top of the band and $\lambda \rightarrow \pm\infty$ at its two ends. Right: the two-spinon continuum, with the exact levels of the $S^z = 1$ sector at $N = 16$ and $N = 18$ that fall inside it. The window is an energy filter, not a classification: at these sizes a few genuine two-hole levels lie just above the dashed thermodynamic boundary and are filtered out along with the four-spinon states.

The continuum, and what a neutron sees. A two-spinon state has $\omega = \varepsilon(p_1) + \varepsilon(p_2)$, $k = p_1 + p_2$, with $p_1, p_2 \in (0, \pi)$ distinct (the holes sit at distinct vacancies). At fixed k , $\sin p_1 + \sin p_2 = 2 \sin \frac{k}{2} \cos \frac{p_1 - p_2}{2}$ grows as $p_1 - p_2$ shrinks and falls as one spinon reaches an end of its band.

Corollary 17.12 (The two-spinon continuum). *The two-spinon states are dense in*

$$\frac{\pi J}{2} |\sin k| \leq \omega \leq \pi J |*| \sin \frac{k}{2}, \quad 0 < k < 2\pi, \quad (17.13)$$

and fill its interior. Neither boundary is attained: the upper one is the limit $p_1 \rightarrow p_2$, the lower one the limit in which one spinon runs to the end of its band, $\lambda \rightarrow \pm\infty$.

The lower boundary is the des Cloizeaux–Pearson dispersion of 1962, read for thirty years as *the* spin-wave branch; the upper one shows it is only the edge of a continuum. That is the measurable difference between a spin chain and a magnet: a three-dimensional antiferromagnet scatters neutrons into a sharp magnon line, the spin- $\frac{1}{2}$ chain into a broad continuum, because a neutron flips one spin and therefore makes spinons in *pairs* — two at least, any even number in principle — which share the energy as they like. Inelastic scattering on KCuF_3 sees such a continuum, and the 1995 measurement was fitted with an

approximate two-spinon cross-section, the Müller ansatz. The exact two-spinon contribution, computed later, exhausts about 73% of the integrated intensity; four spinons and more carry the rest · cited: Tennant, Cowley, Nagler and Tsvelik, *Phys. Rev. B* **52** (1995) 13368; Müller, Thomas, Beck and Bonner, *ibid.* **24** (1981) 1429; Lake et al., *Phys. Rev. Lett.* **111** (2013) 137205.

Which way does p run? Theorem 17.11 could as well have printed $p = 2\pi \int_{\lambda}^{\infty} \rho = \frac{\pi}{2} - \arctan \sinh \pi \lambda$: it satisfies $\varepsilon = \frac{\pi J}{2} \sin p$ just as well, and for a pair of holes at $\pm \lambda$ gives the same total, π . A symmetric test cannot tell the two apart — which is exactly why one must test an asymmetric pair. Against finite chains with the holes placed asymmetrically the printed orientation is right to 10^{-4} at $N = 256$ and the other is wrong by 1.13. The counterexample lives in the check script.

Reading

17.11 Two limits of the entropy, the filling fraction, and the experiment

Remark 17.13 (Two familiar limits of one formula). $\vartheta = 0$ and $\vartheta = 1$ both give zero: an empty region and a full one each admit one arrangement. The maximum is $\ln 2$ per slot at $\vartheta = \frac{1}{2}$. And (17.4) is the entropy of a set of *fermionic* modes — one slot, one particle — even though the gas of Lecture 11 is made of bosons. The exclusion is not in the statistics of the particles but in the Bethe equations, which forbid two equal quantum numbers.

Equation (17.7) is not a model equation of state; it is the equation of state of a real gas. Ultracold ^{87}Rb on an atom chip, confined so tightly in two directions that only the lowest transverse mode is populated, is the Lieb–Liniger gas of Lecture 11 with c set by the transverse confinement. Measuring the *in situ* density profile of such a cloud, and reading the trap’s slowly varying potential as a local chemical potential $\mu(x) = \mu_0 - V(x)$, turns one image into a curve of n against μ at fixed T . Van Amerongen and co-workers did this in 2008 for a weakly interacting gas on an atom chip, fitting only μ_0 and T : no interaction parameter is free, since c follows from the transverse confinement. One complication is instructive — at their temperatures the higher transverse modes are not empty, so what is fitted is $n_{\text{YY}}(\mu(x), T) + \sum_{j \geq 1} (j+1)n_{\text{ideal}}(\mu(x) - j\hbar\omega_{\perp}, T)$: Yang–Yang for the one-dimensional gas, an ideal gas for everything above it. With that, the profiles are reproduced across the crossover from the quasi-condensate to the nearly ideal gas — where mean-field theory has nothing to say · cited: van Amerongen et al., *Phys. Rev. Lett.* **100** (2008) 090402.

Reading

17.12 Strings, and the chain’s thermodynamics

Everything so far used only real roots, and for the gas that is a theorem (Proposition 11.4). The chain is different. Lecture 10 already met a solution with $\lambda_{1,2} = x \pm \frac{i}{2} + O(\delta)$ and δ exponentially small in N (10.23), and its argument — an imaginary part forces a nearly vanishing denominator — applies to any number of roots.

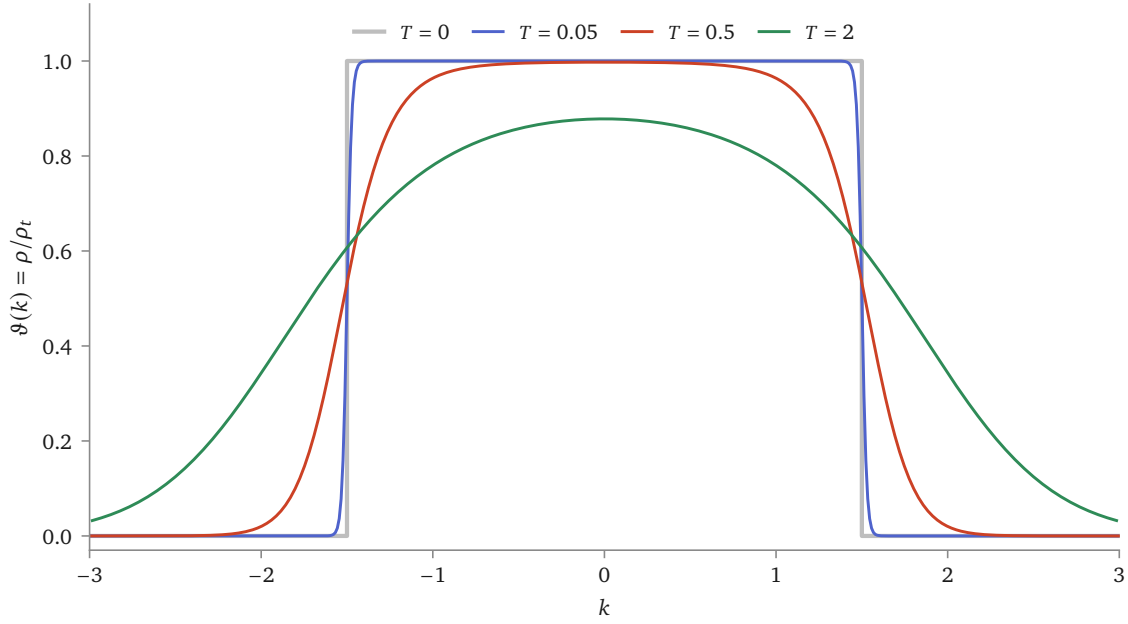


Figure 17.6: The filling fraction $\vartheta = \rho/\rho_t$ for the same three temperatures, against the step function of the $T = 0$ sea. On this axis the crossing region has width $\sim T/|\epsilon'|$ — it widens by a factor ten from $T = 0.05$ to $T = 0.5$ — which is T/v^{eff} only after converting rapidity to dressed momentum, $p' = 2\pi\rho_t$. By $T = 2$ the sea is no longer degenerate at all: $\vartheta < 0.88$ everywhere. ϑ is the one object that carries the state, at $T = 0$ (Lecture 16), here, and out of equilibrium (Lecture 20).

Definition 17.14 (Strings, and the hypothesis). An n -string is a set of n roots $\lambda_\alpha^n + \frac{i}{2}(n+1-2a)$, $a = 1, \dots, n$, with real centre λ_α^n . The *string hypothesis* is that as $N \rightarrow \infty$ the roots of the solutions that matter thermodynamically organise into strings, up to deviations exponentially small in N .

· cited, and false in detail: Essler, Korepin and Schoutens (1992)

The hypothesis is known to fail. There are “wide pairs” whose imaginary parts are not $\pm \frac{1}{2}$, and 2-strings that collapse onto the real axis as N grows; Lecture 16 met the tail of the same phenomenon in the pair of real roots sharing a quantum number at $N = 22$ (Remark 16.2). What survives — and what the thermodynamics needs — is the *counting*: the string configurations reproduce the right number of states even where individual solutions are wrong. That is less than it sounds, and this reading returns to it: a partition function weights states by *energy*, so equal cardinality is not equal free energy. This is an uncomfortable position to be in and it is the honest one.

Bethe–Takahashi. Put the string of Definition 17.14 into the Bethe equations and multiply the n equations of its members together. The exponentially small deviations cancel between a diverging and a vanishing factor, and what is left is one equation per string *centre*:

$$N\theta_n(\lambda_\alpha^n) - \sum_{m,\beta} \Theta_{nm}(\lambda_\alpha^n - \lambda_\beta^m) = 2\pi I_\alpha^n, \quad \theta_n(\lambda) = 2 \arctan \frac{2\lambda}{n}, \quad (17.14)$$

with the scattering phase between an n -string and an m -string

$$\Theta_{nm} = \theta_{|n-m|} + 2\theta_{|n-m|+2} + \dots + 2\theta_{n+m-2} + \theta_{n+m}, \quad \theta_0 \equiv 0. \quad (17.15)$$

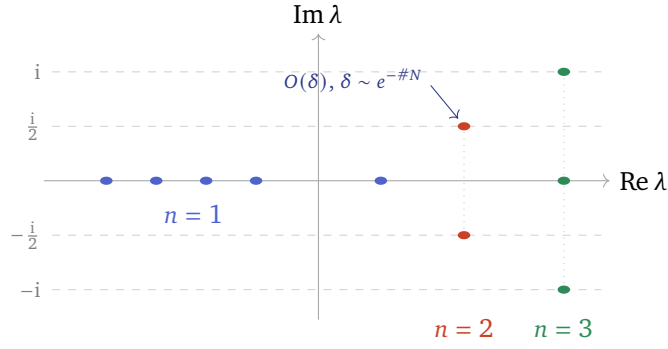


Figure 17.7: The string hypothesis: every root belongs to a string $\lambda^n + \frac{i}{2}(n+1-2a)$, $a = 1, \dots, n$, centred on the real axis and symmetric about it, up to deviations exponentially small in N — so a 2-string sits at $\pm \frac{i}{2}$ and a 3-string at $0, \pm i$. Real roots are 1-strings; an n -string is a bound state of n magnons. It is the *centres* λ^n that the thermodynamics treats as particles.

· *sketched: the cancellation of the deviations is the content, and it is cited (Takahashi 1971)* For $n = m = 1$ this is $\Theta_{11} = \theta_2$ and (17.14) is Lecture 16's (16.1). Two consequences follow as they did there, by reading off the values at $\pm\infty$: since $\Theta_{nm}(\pm\infty) = \pm\pi t_{nm}$ with $t_{nm} = 2 \min(n, m) - \delta_{nm}$, the number of vacancies for n -strings is

$$\#\{\text{vacancies for } n\text{-strings}\} = N - \sum_m t_{nm} M_m, \quad \rho_n + \rho_n^h = a_n - \sum_m A_{nm} * \rho_m, \quad A_{nm} = \frac{\Theta'_{nm}}{2\pi}, \quad (17.16)$$

with M_m the number of m -strings and $a_n = \theta'_n/2\pi$ as in Theorem 16.3. The energy follows from the identity $2 \sum_{a=1}^n (4\lambda_a^2 + 1)^{-1} = \pi a_n(\lambda)$ for an exact string, together with Lecture 10's $E = -\sum_j 2J/(4\lambda_j^2 + 1)$ (10.16): an n -string of centre λ carries energy $-\pi J a_n(\lambda)$ and S^z lowered by n . For $n = 1$ that is $\varepsilon_0 = -\pi J a_1$, the bare energy § 17.5 asked for.

The chain's TBA, and the Y-system. Repeat § 17.5 with one density per string length. Put the chain in a field, $H \mapsto H - h \sum_n S_n^z$. An n -string lowers S^z by n , so the Zeeman energy is $-hN/2 + h \sum_n n M_n$: the magnons' chemical potential is $-h$, and it is $+h$ that takes the place of the gas's $-\mu$. Nothing else changes except the bookkeeping, and the answer is an infinite coupled system: with $\eta_n = e^{\varepsilon_n/T} = \rho_n^h/\rho_n$,

$$\varepsilon_n = -\pi J a_n + hn + T \sum_m A_{nm} * \ln(1 + \eta_m^{-1}), \quad f = -\frac{h}{2} - T \sum_n \int a_n \ln(1 + \eta_n^{-1}) d\lambda. \quad (17.17)$$

· *sketched: this is Theorem 17.7's variation with one density per string length, not redone here*

The sign in front of the sum has flipped because $K_{XX}/2\pi = -a_2$ (Lecture 16): at $n = m = 1$, (17.17) is (17.7). Infinitely many equations, each with infinitely many terms, is not progress. It becomes progress because the matrix $1 + A$ has an inverse that is almost diagonal: in Fourier variables $(1 + A)^{-1} = 1 - \hat{s}$ (shift in n) with $\hat{s}(\omega) = (2 \cosh(\omega/2))^{-1}$, i.e. with the kernel $s(\lambda) = (2 \cosh \pi \lambda)^{-1}$. Applying it collapses (17.17) to

Theorem 17.15 (Takahashi's TBA for the Heisenberg chain). With $\eta_0 \equiv 0$ and $s(\lambda) =$

$$(2 \cosh \pi \lambda)^{-1},$$

$$\ln \eta_n = -\delta_{n,1} \frac{\pi J}{T} s + s^* \ln[(1 + \eta_{n-1})(1 + \eta_{n+1})], \quad f = -J \ln 2 - T \int s(\lambda) \ln(1 + \eta_1(\lambda)) d\lambda, \quad (17.18)$$

with $h \geq 0$ entering only through $\lim_{n \rightarrow \infty} (\ln \eta_n)/n = h/T$.

• *sketched: the Fourier inversion is checked in the script; the convergence of the truncation is numerical*

Three things fall out at once. The driving term now sits at $n = 1$ alone, and every other equation couples only to its two neighbours — a chain in the index n . The constant $-J \ln 2$ in f is [Lecture 16](#)'s Hulthén ground-state energy (16.9), and it arrives here as $\int s \varepsilon_0 = -J \int s \pi a_1 = -J \ln 2$, not by assumption. And at $T \rightarrow \infty$ the driving vanishes and (17.18) has the constant solution $\eta_n = n(n+2)$, whence $f \rightarrow -T \int s \ln 4 = -T \ln 2$ — the free energy of N free spins, by a telescoping product.

Remark 17.16 (The Y-system). s is the inverse of $g \mapsto g(\lambda + \frac{i}{2}) + g(\lambda - \frac{i}{2})$, and $s(\lambda \pm \frac{i}{2}) = \mp (2i \sinh \pi \lambda)^{-1}$, whose sum vanishes identically. Those two facts use the same symbol for two prescriptions — approached from inside the strip the shifted sum of s is a nascent δ , which is exactly why the operator inverts s^* — so the driving term is not destroyed but *re-encoded*. For η_n analytic in $|\operatorname{Im} \lambda| \leq \frac{1}{2}$, applying the operator to (17.18) away from $\lambda = 0$ leaves

$$\eta_n(\lambda + \frac{i}{2}) \eta_n(\lambda - \frac{i}{2}) = (1 + \eta_{n-1}(\lambda))(1 + \eta_{n+1}(\lambda)). \quad (17.19)$$

The temperature, the coupling and the field have vanished: they survive in the analyticity and the asymptotics alone. *The Y-system is universal*, and it is where [Lecture 15](#)'s Q reappears: *fusing* the auxiliary space gives a tower of transfer matrices whose eigenvalues obey the same hierarchy and are generated from a single Q (15.20). *The thermal η_n are not those ratios* — the same functional relation has many solutions, and the temperature is in the analyticity — which is the point the reading makes.

In practice one truncates at some $n_{\max}$ and, at zero field, imposes $\eta_n \rightarrow n(n+2)$ beyond it. At $T = J$ the truncation error falls below 10^{-5} by $n_{\max} = 20$, and the free energy so computed agrees with exact diagonalisation of the finite chain — with a *finite* chain, where no string ever appears — to five decimals at $T = 2J$, J and $J/2$. The check script does exactly that comparison, and it is the strongest evidence *this reading offers* that the string hypothesis, false in detail, counts correctly — not the strongest there is, since the same equations follow algebraically from the quantum transfer matrix with no string hypothesis at all • *cited: Essler et al. (2005)*. Note what it settles. That the string configurations preserve the *number* of states does not by itself give the free energy, since Z weights states by energy; it is the agreement of the *energies* that is the evidence.

17.13 The field's factor, and where Q reappears

Two things § 17.12 asserted are worth doing properly: the field's normalisation, which no check of the equations can see, and the promise Lecture 15 made about the T-Q relation, which is worth keeping exactly as far as it is true.

The field. Everything in (17.18) except the boundary condition is blind to h : the operator that produced it annihilates the sequence hn (check script). So the normalisation of h survives in exactly one place, and it is the one place no check of the equations can see. Set $J = 0$, where the answer is known. The recurrence has the constant solution $1 + \eta_n = (\sinh((n+1)x)/\sinh x)^2$, for which $(\ln \eta_n)/n \rightarrow 2x$, and (17.18) gives $f = -T \ln(2 \cosh x)$. A free spin in $H - h \sum S^z$ has $f = -T \ln(2 \cosh(h/2T))$, so $x = h/2T$ and the boundary condition is h/T . Takahashi's is $2h_T/T$ because his field term is $-2h_T \sum S^z$: taking his boundary condition with our Hamiltonian multiplies the zero-field susceptibility by four. As $x \rightarrow 0$ the hyperbolic solution becomes $\eta_n = n(n+2)$, which is why that is the zero-field truncation.

Where Q reappears, and where it does not. Fusing the auxiliary space of Lecture 14 to spin $\frac{n}{2}$ gives a tower of transfer matrices T_n , and their eigenvalues obey a T -system: a relation of the shape

$$T_n^{[1]} T_n^{[-1]} = T_{n-1} T_{n+1} + f_n, \quad g^{[m]}(\lambda) = g\left(\lambda + \frac{im}{2}\right), \quad (17.20)$$

with f_n a known scalar built from the vacuum factors. Put $Y_n = T_{n-1} T_{n+1} / f_n$; then (17.20) at $n \mp 1$ gives $1 + Y_{n \mp 1} = T_{n \mp 1}^{[1]} T_{n \mp 1}^{[-1]} / f_{n \mp 1}$, and the Y_n satisfy (17.19) provided $f_n^{[1]} f_n^{[-1]} = f_{n-1} f_{n+1}$ — a condition on the scalar alone, which the standard f_n satisfies identically (check script). And every fused eigenvalue is generated from a single Baxter Q together with those vacuum factors: that is the sense in which Lecture 15's T-Q relation (15.20) generalises · cited: Kuniba, Nakanishi and Suzuki (2011) §§6, 8.

Now the warning, which is the reason this is a reading and not a theorem. The Y_n so constructed are **not** the thermal η_n . They cannot be: an ordinary transfer matrix of a finite chain has no temperature in it. At $N = 2$ on the reference state the ratio $T_0 T_2 / f_1$ is a non-constant rational function of λ , negative on an interval, where the $J = h = 0$ thermal solution is the constant $\eta_1 = 3$. A functional equation does not determine its solution; the temperature enters through analyticity and asymptotics, and the object whose ratios are the thermal η_n is the quantum transfer matrix, a Trotter-limit construction this course does not build · cited: Essler et al. (2005), §13.2 and appendix 13.A. What Lecture 15 promised is delivered here at the level of the hierarchy — the Y-system of (17.19) is the T-Q relation's, seen from the other side — and not at the level of the solutions.

17.14 Why it works, and what it does not prove

Theorem 17.7 minimises a functional. What entitles anyone to believe that the minimum is the free energy?

The saddle-point argument is in § 17.5.

Yang and Yang proved what can be proved cleanly: that (17.7) has a unique solution for every $c > 0$, $T > 0$, that ε is analytic in μ and T and increasing in $|k|$, and that the resulting $P(\mu, T)$ is convex — so that the thermodynamics it defines is consistent · cited: Yang and Yang (1969). What their argument does *not* establish is the step before: that the set of Bethe

states really is in bijection with the sets of quantum numbers, and that the coarse-graining loses nothing. For the gas, half of the first is a theorem: quantum numbers and solutions are in bijection (Proposition 11.5); that the states so built are *all* of them is still cited. For the chain even that much rests on the string hypothesis, which is false as a statement about individual solutions. The subject's position is that the counting is right even so. Part of the evidence is numerical — § 17.12's exact diagonalisation, at the three temperatures it reports, and no further, since a ring of fourteen sites is not the thermodynamic limit at low temperature. The rest is theoretical, and independent: the same equations come out of the quantum transfer matrix without any string hypothesis · cited: Essler et al. (2005), §13.2 and appendix 13.A.

There is a second, sharper reason to trust it. The same ϑ that minimises g is the one that appears in the dressing equation, and Corollary 17.8 makes the dressed velocity a derivative of the free energy's own ingredients. Thermodynamics and dynamics are answered by one function. Lecture 20 takes that seriously enough to let ϑ depend on position and time.

One more caution. ε is not a *bare, state-independent* one-particle energy: it depends on T and μ , and only at $c = \infty$ does it reduce to $k^2 - \mu$. What it *is* is the *dressed* cost of a particle-hole excitation above the thermal state — moving a root from λ_h to λ_p costs $\varepsilon(\lambda_p) - \varepsilon(\lambda_h)$, backflow included · cited: Korepin, Bogoliubov and Izergin §1.8 — which at $T = 0$ reduces to the dressed energy (17.3). The trap is the next step: reading $e^{-\varepsilon/T}$ as a Boltzmann weight for a particle of energy ε gets the filling right for the wrong reason, and stops working the moment one asks for a correlation, where the shape of the state and not only its occupations is needed.

Tutorial

17.15 Solving the Yang–Yang equation

The recipe is Lecture 16's Nyström method (16.15) wrapped in the iteration of Proposition 17.10. Take an n -point Gauss–Legendre rule on $[-\Lambda, \Lambda]$, start from $\varepsilon = k^2 - \mu$, and iterate

$$\varepsilon_i \leftarrow k_i^2 - \mu - \frac{T}{2\pi} \sum_j w_j K(k_i - k_j) \ln(1 + e^{-\varepsilon_j/T}), \quad (17.21)$$

then read off P , ϑ , and — by one linear solve with the same matrix — ρ_t and every dressed quantity. Three practical points, all of which cost something before they were learnt.

The cutoff and the grid answer to different scales. Λ is set by where $\ln(1 + e^{-\varepsilon/T})$ dies — by $\sqrt{\mu + \#T}$, and read off the *interacting* ε , not the bare one. The number of nodes is set by the sharpest feature, and there are two: at low temperature the Fermi edge, of width $T/|\varepsilon'|$ in rapidity (*not* T/ν^{eff} , which is its width in dressed momentum), and at small c the kernel itself, a peak of width c . Large c is therefore the *easy* end — the kernel flattens towards $2/c$ — and small c is the *hard* one, where a rule that does not resolve the peak has a Nyström diagonal $w_i/\pi c$ that diverges instead of tending to a row integral of one. What is not set by c is Λ : taking $\Lambda \propto c$, as looked natural, put the nodes so far apart that the $c = 100$ pressure came out as 10^{-6} .

Convergence in n is not uniform in T . At $c = \mu = 1$ with the cutoff $\Lambda = \sqrt{\mu + 60T} + 12$ used throughout (so $\Lambda = 16.0$ at $T = 0.25$ and 13.3 at $T = 0.01$), an 800-point rule is converged to ten digits at $T = 0.25$, where 200 points give 3×10^{-5} ; at $T = 0.01$ the same 200 points are wrong in the *fourth* digit, for exactly the reason above. *Quote a cutoff with every such number* — at $\Lambda = 20$ the warm 200-point error is six times larger. As in Lecture 16: run n

against $2n$, and do it at the temperature you mean to use, not at the one where the code was written.

Check the thermodynamics, not the equation. The strongest test costs two extra solves: n against $\partial P/\partial \mu$ and s against $\partial P/\partial T$, with s computed from (17.4) and n from $\int \rho$. These agree to eight digits when everything is right and fail loudly when a factor of 2π is not. They also confirm the equation's weakest link — ρ_t comes from a linear solve that never appears in (17.7), so nothing else would catch an error in it.

A worked number. At $c = 1$ and the $\mu = 1.5554$ that (17.3) gives for $q = 1.5$ (so $\gamma = 0.930$, $e = 0.604$), the sound velocity is $v^{\text{eff}} = 1.9086$ and the low-temperature pressure obeys $P(T) - P(0) = \pi T^2/6v^{\text{eff}}$ to four digits at $T = 0.02$: the Sommerfeld correction with central charge one, which is the Luttinger-liquid statement of Lecture 16's reading seen from the thermodynamic side.

Finally, the chain. The same iteration solves (17.18), but now s^* is a convolution best done by fast Fourier transform on a uniform grid, the index n must be truncated, and the initial guess $\eta_n = n(n+2)$ is the right one because it is the exact $T = \infty$ solution. Twenty string lengths and a grid of 2^{12} points on $|\lambda| \leq 40$ give the free energy to five decimals in about six seconds (thirty lengths cost three times that and buy one more decimal), which is enough to put it beside the exact diagonalisation of a 14-site ring and watch the two agree.

Summary.

- *Dressing* (Definition 17.1): $f^{\text{dr}} = f + \frac{1}{2\pi} K \vartheta f^{\text{dr}}$; $2\pi \rho_t = (p'_0)^{\text{dr}}$, and at $T = 0$ with the edge condition $v^{\text{eff}} = (\varepsilon'_0)^{\text{dr}}/(k'_0)^{\text{dr}}$ (Proposition 17.2).
- *Lieb's branches*: a particle above the sea (Type I) and a hole in it (Type II), both linear at small momentum with the speed of sound.
- *Counting* (Proposition 17.4): a state is a choice of filled vacancies, and its entropy is that of fermionic slots.
- *Yang–Yang* (Theorem 17.7): minimising $g = e - \mu n - Ts$ gives $\varepsilon = \varepsilon_0 - \mu - \frac{T}{2\pi} K * \ln(1 + e^{-\varepsilon/T})$, ϑ a Fermi function of ε , and the pressure; dressing holds at every T (Corollary 17.8).
- *Next*: Lecture 18 solves the Ising chain by free fermions, the model Lecture 19 needs out of equilibrium.

Tutorial. Exercise Sheet 17.

References

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